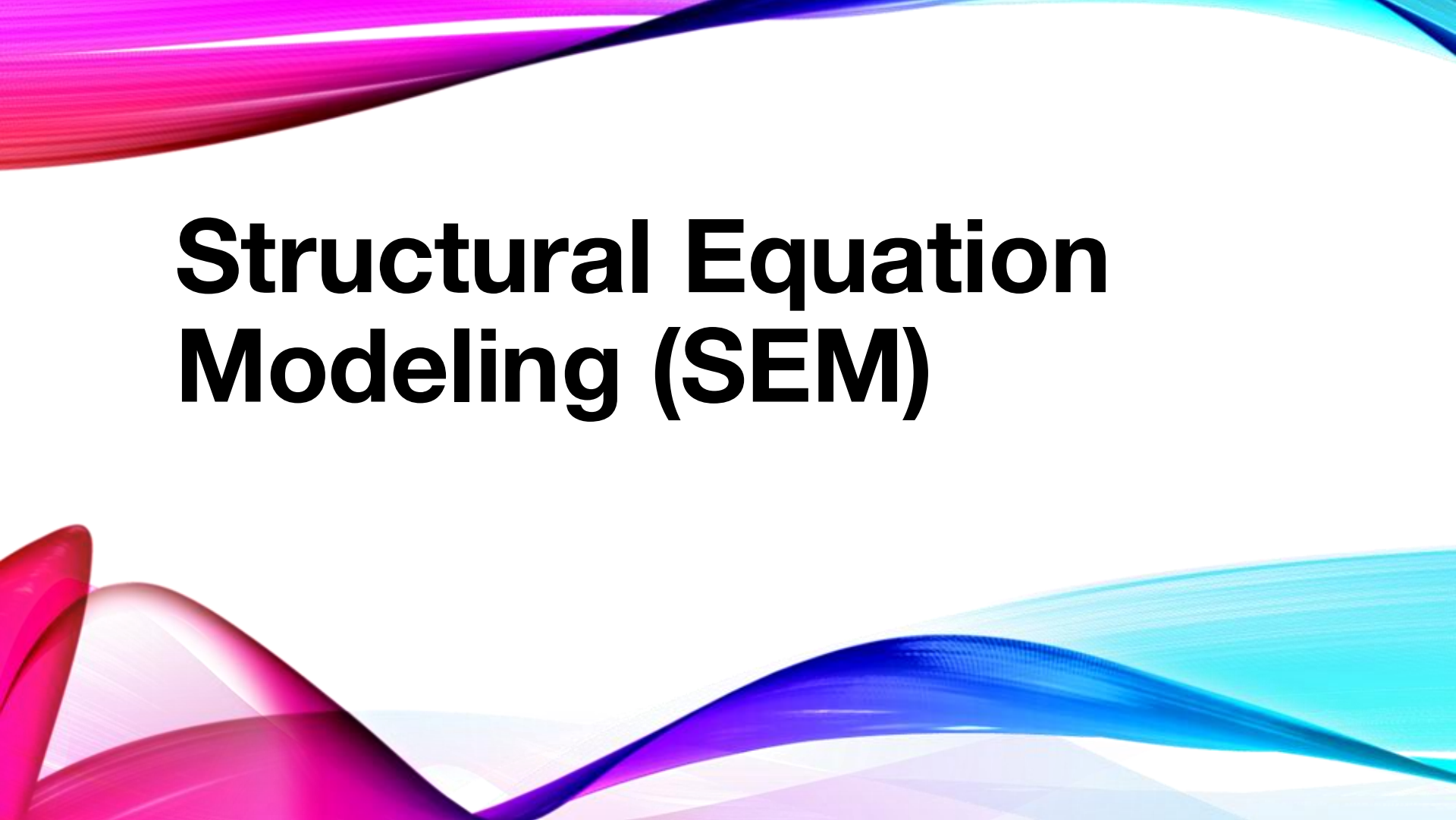




# **Structural Equation Modeling (SEM)**



# **Introduction**

# What is SEM?

- ❖ **Structural equation modeling** (SEM) is a family of statistical procedures
  - Analyze complex multivariate relationships
  - Causal inference method
- ❖ Great for testing **theoretical** variables and getting numerical outputs
  - Takes qualitative, theoretical hypotheses and turns them into numeric estimates
- ❖ Success isn't based on if the model "fits" the data, but if the analysis is able to address those theoretical issues

# SEM: The Origin Story

- ❖ Origins in the social psychology field
  - **Spearman's** factor analysis (1904)
    - Looking at general intelligence underlying test performance
  - **Wright's** path analysis (1918; 1921)
    - Biological and genetic focus
- ❖ SEM is widely used in psychology
  - Beierlein et al. (2016) → “group-focused enmity”
  - Assess several different things in one model

# Example Research Question

- ❖ Industrialization and Political Democracy dataset (Bollen, 1989)
- ❖ Two abstract variables:
  - **Industrialization**
    - x1: Gross National Product per Capita
    - x2: Energy Consumption per Capita
    - x3: Percentage of Labor Force in Industry
  - **Political Democracy**
    - y1: Freedom of the Press
    - y2: Freedom of Political Opposition
    - y3: Fairness of Elections
    - y4: Effectiveness of the Elected Legislature
    - Same 4 measured again in 1965, renamed y5, etc.

# Example Research Question

- 1. Does a country's industrialization level predict its political democracy in 1960?**
- 1. Do political democracy and industrialization in 1960 predict its political democracy in 1965?**



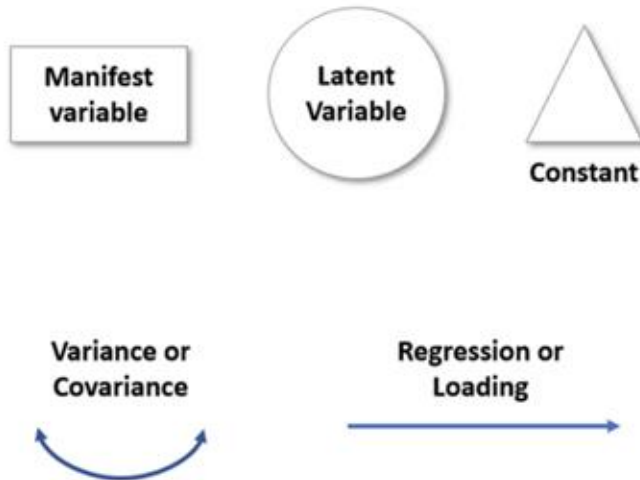
# Theory

# Components of SEM

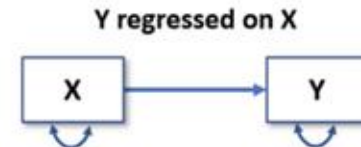
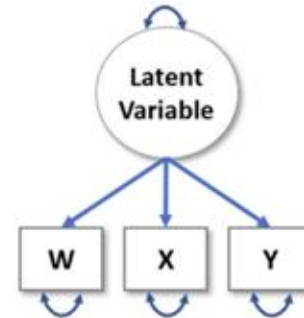
- ❖ **Observed** variables vs. **latent** variables
  - **Observed** → the variables you actually collected data for
  - **Latent** → hypothetical constructs
- ❖ An **indicator** is an observed variable that reflects a latent variable
- ❖ A **factor** is the statistical version of a latent variable
- ❖ Two main structures:
  - **Covariance** → Assesses variances/covariances;  $r_{XY} * SD_X * SD_Y$
  - **Mean** → Estimation of means on latent variables
- ❖ **Structural model** → relationship among the constructs themselves

# Graphing SEM

## SEM Path Diagram Elements



## Fundamental SEM Examples



# SEM Assumptions

- ❖ Multivariate normality
- ❖ Linearity
- ❖ No perfect multicollinearity
- ❖ No systematic missing data
- ❖ Sufficiently large sample size
  - **N:q rule** → Sample size/model parameters ratio; 20:1
- ❖ Correct model specification
  - **Specification error** → omission of relevant variables
- ❖ Independence of observations
- ❖ No extreme outliers

# Strengths & Weaknesses

## Strengths

- ❖ Distinguishes between observed vs. latent variables
- ❖ Address abstract questions
- ❖ Exploratory and confirmatory analyses
- ❖ Extremely powerful
- ❖ Comprehensive and complex analyses

## Weaknesses

- ❖ Super complicated
- ❖ Requires large samples
- ❖ Easy to misinterpret
- ❖ Goodness-of-fit tests can be misleading
- ❖ Inappropriate for nonlinear data
- ❖ Still developing

# When Will I Ever Use This?

## ❖ This is great for...

- Model variables that can't be measured directly
- Model variables with measurement error
- Models where variables are both predictors/outcomes
- Testing theories/concepts
- Handling missing data

## ❖ Potential alternatives:

- Factor score path analysis for small samples (Devlieger & Rosseel, 2017)
- Multilevel modeling for nested data (Mehta & Neale, 2005)



# **R Demonstration**



# **Interpretation**

# Interpretation: Model Fit

lavaan 0.6-21 ended normally after 66 iterations

|                                |        |
|--------------------------------|--------|
| Estimator                      | ML     |
| Optimization method            | NLMINB |
| Number of model parameters     | 31     |
| Number of equality constraints | 3      |
| Number of observations         | 75     |

## Model Test User Model:

|                                         | Standard | Scaled |
|-----------------------------------------|----------|--------|
| Test Statistic                          | 40.179   | 44.492 |
| Degrees of freedom                      | 38       | 38     |
| P-value (Chi-square)                    | 0.374    | 0.217  |
| Scaling correction factor               |          | 0.903  |
| Yuan-Bentler correction (Mplus variant) |          |        |

## Model Test Baseline Model:

|                           |         |         |
|---------------------------|---------|---------|
| Test statistic            | 730.654 | 702.841 |
| Degrees of freedom        | 55      | 55      |
| P-value                   | 0.000   | 0.000   |
| Scaling correction factor |         | 1.040   |

## User Model versus Baseline Model:

|                                    |       |       |
|------------------------------------|-------|-------|
| Comparative Fit Index (CFI)        | 0.997 | 0.990 |
| Tucker-Lewis Index (TLI)           | 0.995 | 0.985 |
| Robust Comparative Fit Index (CFI) |       | 0.991 |
| Robust Tucker-Lewis Index (TLI)    |       | 0.987 |

## Model Significance

| N  | $\chi^2$ | df | p     |
|----|----------|----|-------|
| 75 | 40.179   | 38 | 0.374 |

## Model Fit

| CFI   | RMSEA | 90% CI        | TLI   | SRMR  | AIC      | BIC      |
|-------|-------|---------------|-------|-------|----------|----------|
| 0.997 | 0.028 | 0.000 — 0.087 | 0.995 | 0.056 | 3153.636 | 3218.526 |

Our **Chi-square** is insignificant,  $p = .374$

Our **CFI** is .997, which means our model closely fits the data.

# Interpretation: Model Fit

|                                       |          |          |
|---------------------------------------|----------|----------|
| Akaike (AIC)                          | 3153.636 | 3153.636 |
| Bayesian (BIC)                        | 3218.526 | 3218.526 |
| Sample-size adjusted Bayesian (SABIC) | 3130.277 | 3130.277 |

Root Mean Square Error of Approximation:

|                                               |       |       |
|-----------------------------------------------|-------|-------|
| RMSEA                                         | 0.028 | 0.048 |
| 90 Percent confidence interval - lower        | 0.000 | 0.000 |
| 90 Percent confidence interval - upper        | 0.087 | 0.100 |
| P-value H <sub>0</sub> : RMSEA ≤ 0.050        | 0.665 | 0.492 |
| P-value H <sub>0</sub> : RMSEA ≥ 0.080        | 0.083 | 0.182 |
| Robust RMSEA                                  |       | 0.045 |
| 90 Percent confidence interval - lower        |       | 0.000 |
| 90 Percent confidence interval - upper        |       | 0.093 |
| P-value H <sub>0</sub> : Robust RMSEA ≤ 0.050 |       | 0.525 |
| P-value H <sub>0</sub> : Robust RMSEA ≥ 0.080 |       | 0.132 |

Standardized Root Mean Square Residual:

|      |       |       |
|------|-------|-------|
| SRMR | 0.056 | 0.056 |
|------|-------|-------|

Model Significance

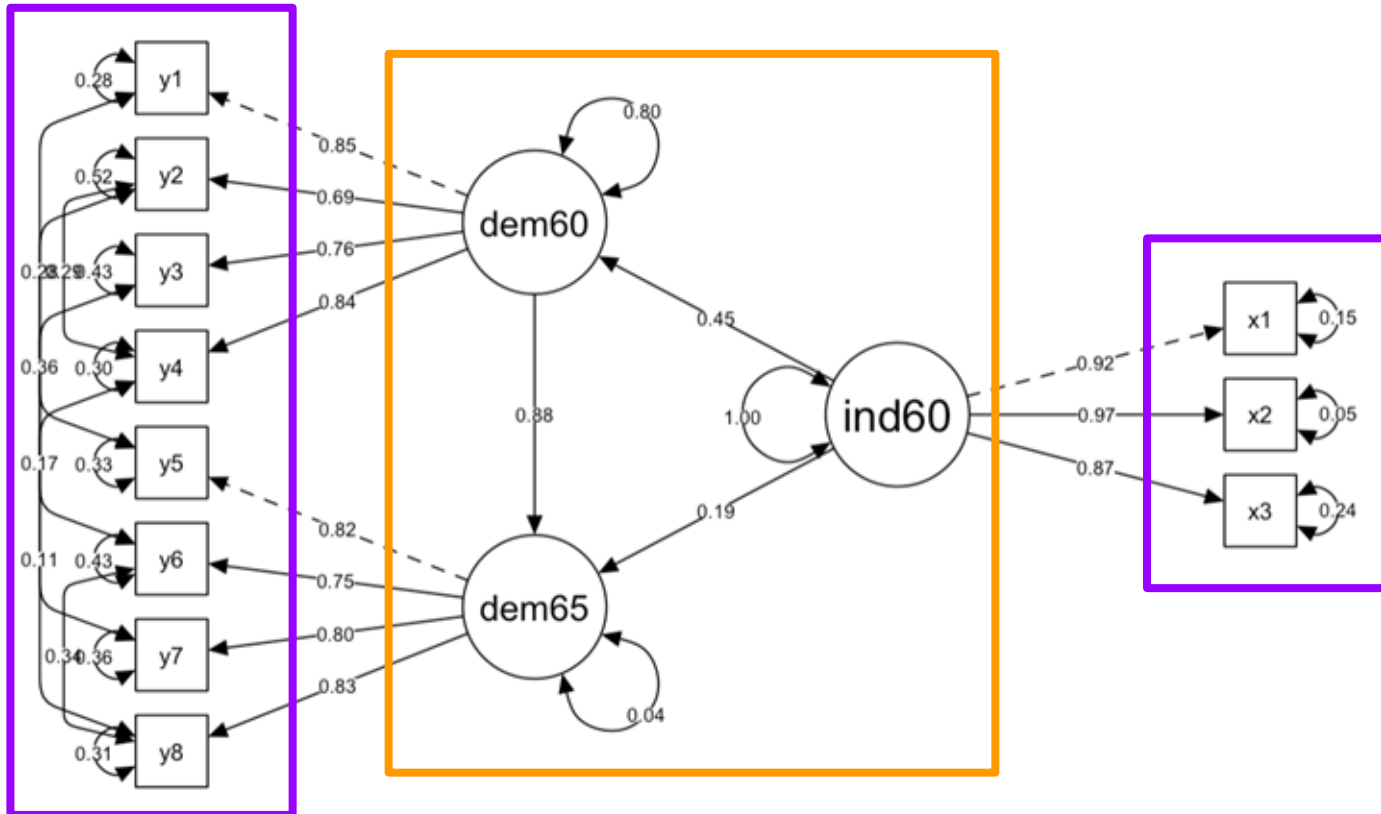
| N  | $\chi^2$ | df | p     |
|----|----------|----|-------|
| 75 | 40.179   | 38 | 0.374 |

Model Fit

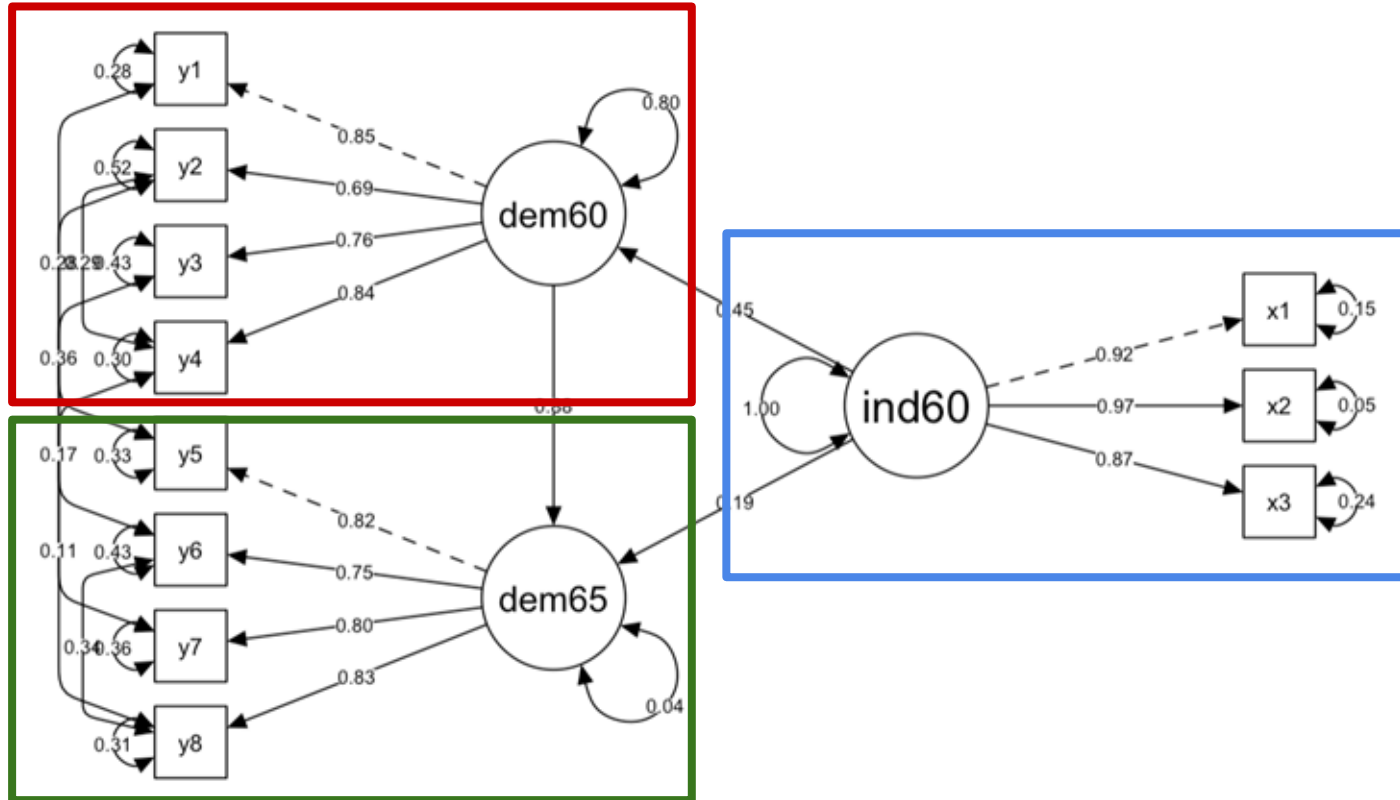
| CFI   | RMSEA | 90% CI        | TLI   | SRMR  | AIC      | BIC      |
|-------|-------|---------------|-------|-------|----------|----------|
| 0.997 | 0.028 | 0.000 — 0.087 | 0.995 | 0.056 | 3153.636 | 3218.526 |

Other **fit indices** indicate that our model fits the data!

# Model Visualization



# Model Visualization



# Interpretation: Factor Loadings

Latent Variables:

|       |     | Estimate | Std.Err | z-value | P(> z ) | Std.lv | Std.all |
|-------|-----|----------|---------|---------|---------|--------|---------|
| ind60 | ==~ |          |         |         |         |        |         |
| x1    |     | 1.000    |         |         |         | 0.670  | 0.920   |
| x2    |     | 2.180    | 0.145   | 15.072  | 0.000   | 1.460  | 0.973   |
| x3    |     | 1.818    | 0.140   | 12.953  | 0.000   | 1.218  | 0.872   |
| dem60 | ==~ |          |         |         |         |        |         |
| y1    |     | 1.000    |         |         |         | 2.201  | 0.850   |
| y2    | (a) | 1.191    | 0.141   | 8.429   | 0.000   | 2.621  | 0.690   |
| y3    | (b) | 1.175    | 0.118   | 9.951   | 0.000   | 2.586  | 0.758   |
| y4    | (c) | 1.251    | 0.138   | 9.067   | 0.000   | 2.754  | 0.838   |
| dem65 | ==~ |          |         |         |         |        |         |
| y5    |     | 1.000    |         |         |         | 2.154  | 0.817   |
| y6    | (a) | 1.191    | 0.141   | 8.429   | 0.000   | 2.565  | 0.755   |
| y7    | (b) | 1.175    | 0.118   | 9.951   | 0.000   | 2.530  | 0.802   |
| y8    | (c) | 1.251    | 0.138   | 9.067   | 0.000   | 2.694  | 0.829   |

Factor Loadings

| Latent Factor | Indicator | Standardized |               |     |       |        |       |
|---------------|-----------|--------------|---------------|-----|-------|--------|-------|
|               |           | Loading      | 95% CI        | sig | SE    | z      | p     |
| dem60         | y1        | 0.850        | 0.763 — 0.938 | *** | 0.044 | 19.114 | 0.000 |
| dem60         | y2        | 0.690        | 0.566 — 0.813 | *** | 0.063 | 10.959 | 0.000 |
| dem60         | y3        | 0.758        | 0.666 — 0.850 | *** | 0.047 | 16.128 | 0.000 |
| dem60         | y4        | 0.838        | 0.748 — 0.927 | *** | 0.046 | 18.376 | 0.000 |
| dem65         | y5        | 0.817        | 0.716 — 0.918 | *** | 0.051 | 15.867 | 0.000 |
| dem65         | y6        | 0.755        | 0.659 — 0.850 | *** | 0.049 | 15.513 | 0.000 |
| dem65         | y7        | 0.802        | 0.721 — 0.882 | *** | 0.041 | 19.524 | 0.000 |
| dem65         | y8        | 0.829        | 0.730 — 0.928 | *** | 0.050 | 16.417 | 0.000 |
| ind60         | x1        | 0.920        | 0.876 — 0.964 | *** | 0.022 | 41.435 | 0.000 |
| ind60         | x2        | 0.973        | 0.941 — 1.005 | *** | 0.016 | 59.451 | 0.000 |
| ind60         | x3        | 0.872        | 0.816 — 0.929 | *** | 0.029 | 30.206 | 0.000 |

\* p < .05; \*\* p < .01; \*\*\* p < .001

**Our factor loadings** are all strong and significant, so they do a good job indicating our latent variables

# Interpretation: Regressions

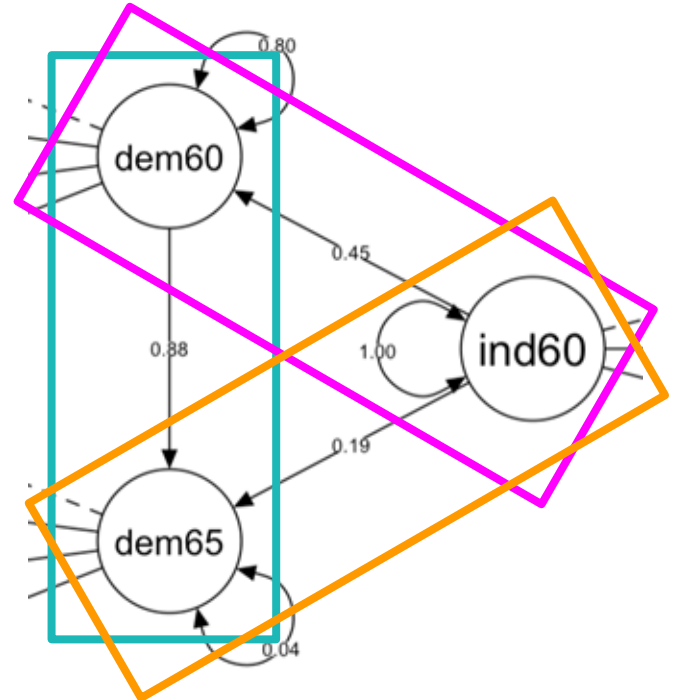
Regressions:

|         | Estimate | Std.Err | z-value | P(> z ) | Std.lv | Std.all |
|---------|----------|---------|---------|---------|--------|---------|
| dem60 ~ |          |         |         |         |        |         |
| ind60   | 1.471    | 0.352   | 4.184   | 0.000   | 0.448  | 0.448   |
| dem65 ~ |          |         |         |         |        |         |
| ind60   | 0.600    | 0.230   | 2.613   | 0.009   | 0.187  | 0.187   |
| dem60   | 0.865    | 0.062   | 14.035  | 0.000   | 0.884  | 0.884   |

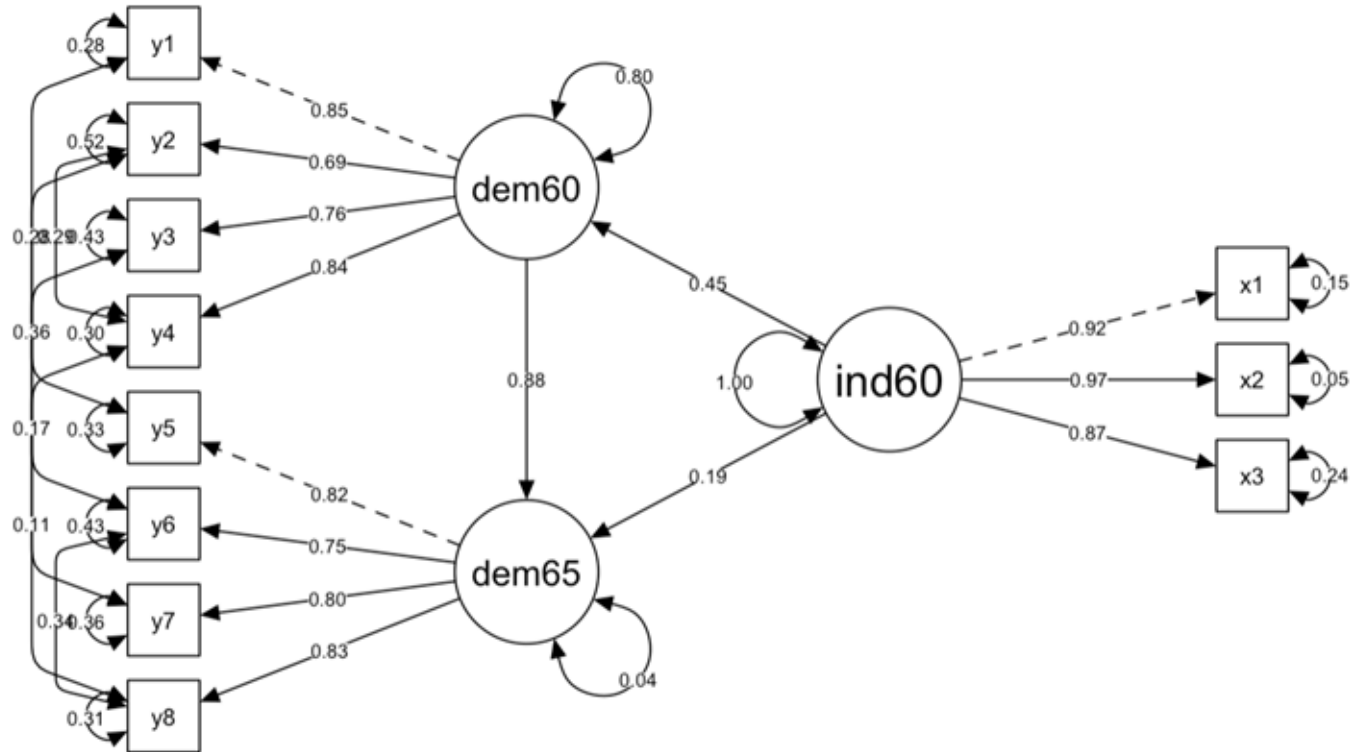
Regression Paths

| Predictor | DV    | Standardized |               |     |       |        |        |
|-----------|-------|--------------|---------------|-----|-------|--------|--------|
|           |       | $\beta$      | 95% CI        | sig | SE    | z      | p      |
| ind60     | dem60 | 0.448        | 0.231 — 0.665 | *** | 0.111 | 4.046  | <0.001 |
| dem60     | dem65 | 0.884        | 0.787 — 0.981 | *** | 0.050 | 17.826 | 0.000  |
| ind60     | dem65 | 0.187        | 0.056 — 0.318 | **  | 0.067 | 2.795  | 0.005  |

\* p < .05; \*\* p < .01; \*\*\* p < .001



# Model Visualization



# Reporting SEM Results

The SEM model was evaluated and demonstrated a good fit to the data. The Chi-square test was not significant,  $\chi^2(38) = 40.18, p = .374$ . The CFI was .997, which indicates the model fit the data. Analysis of the regression paths showed that a country's industrialization level in 1960 significantly predicted its political democracy in 1960,  $\beta = .448, SE = .111, p < .001$ , indicating higher industrialization was associated with higher political democracy scores.

Moreover, a country's political democracy in 1960 significantly predicted its political democracy in 1965,  $\beta = .884, SE = .050, p < .001$ . Political democracy in 1965 was also significantly predicted by industrialization in 1960,  $\beta = .187, SE = .067, p = .005$ . These results indicate that political democracy level (1960) is associated with later democratic development (1965). Results also suggest that industrialization is generally associated with democratic development.

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